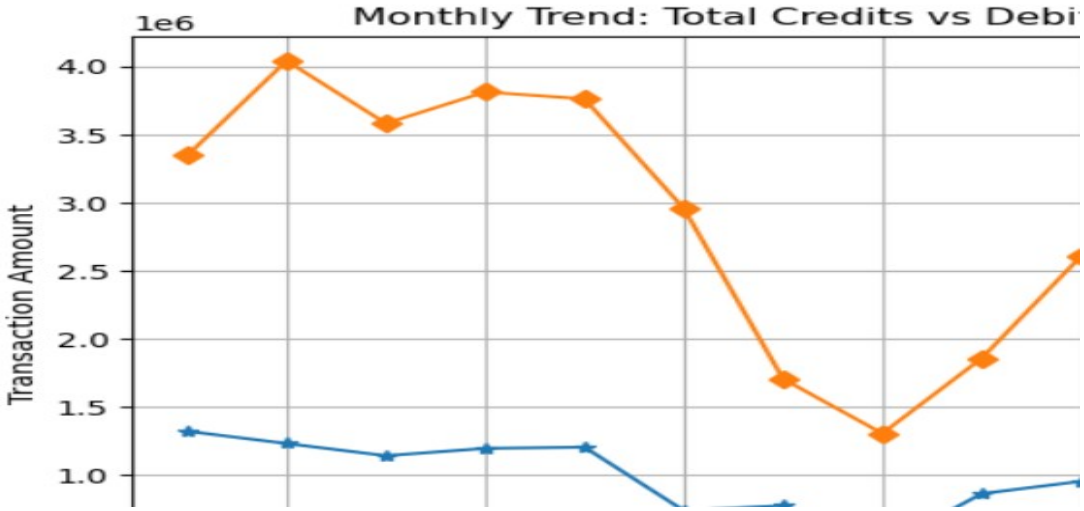
Financial Risk Analysis with Python

Task 1: Data Cleaning and Formatting :

- **Remove / treat special characters or non-numeric entries:** Financial columns were checked using isnull() and info() to confirm there were no special characters, non-numeric values, or missing data.
- **Convert currency amounts into numerical format:** TransactionAmount and AccountBalance were validated to be in numeric format and confirmed as suitable for calculations.
- **Validate and format date columns:** The TransactionDate column was converted to a proper datetime format to enable accurate time-based analysis.
- **Ensure account types and transaction categories are standardized:** AccountType and TransactionType columns were reviewed and validated to ensure consistent and uniform category values.

Task 2: Descriptive Transactional Analysis :

- Monthly and yearly summaries of total credits and debits were calculated using transaction dates. Deposits were treated as credit transactions, while withdrawals, payments, and transfers were treated as debits. Net transaction volume was computed as the difference between total credits and total debits. This analysis helps identify spending trends and financial stability over time.
- A line chart was used to show monthly changes in total credits and debits. Credits show money coming into accounts, and debits show money going out. The chart shows that debits are higher than credits in most months, which means customers are spending more money over time.



- Net inflow was calculated for each account by subtracting total debit amounts from total credit amounts. Accounts with the highest positive net inflow were identified as top performing accounts, while accounts with the highest negative net inflow were identified as bottom performing accounts. This helps highlight financially stable and high-risk accounts.
- Accounts were flagged as dormant if there was a gap of two months or more between consecutive transactions. Transaction dates were sorted by account, and the time difference between transactions was calculated. Accounts with any transaction gap greater than or equal to two months were classified as inactive.

Task 3: Customer Profile Building :

- Accounts were grouped into High, Medium, and Low activity categories based on transaction frequency. Accounts with more than 15 transactions were classified as High activity, those with 5 to 15 transactions as Medium activity, and those with fewer than 5 transactions as Low activity. This classification helps identify highly engaged customers as well as low-usage or inactive accounts.
- Customers were segmented using their average account balance and average transaction volume. Percentile-based thresholds were applied to classify customers into high, medium, and low balance groups, as well as high, medium, and low transaction volume groups. This segmentation helps identify high-value customers, moderate users, and customers with lower financial engagement.
- Customer accounts were profiled based on net inflow, transaction frequency, and average balance. High-net inflow accounts represent financially strong customers, high-frequency low-balance accounts indicate active but potentially risky behavior, and negative or near-zero balance accounts highlight financially stressed customers requiring attention.

Task 4: Financial Risk Identification :

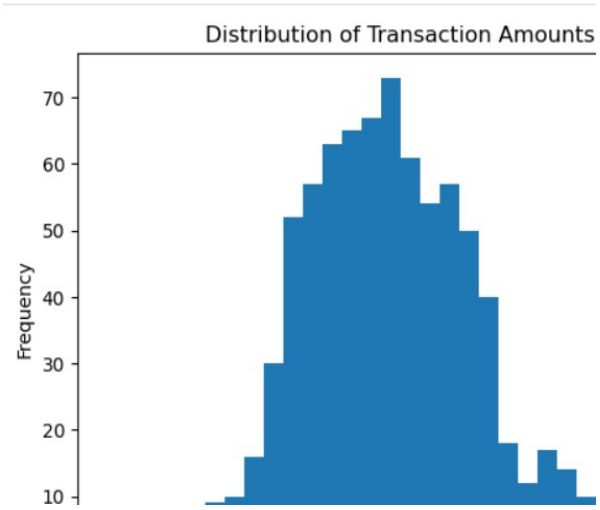
- Accounts were monitored for frequent large withdrawals and overdraft behavior. Large withdrawals were identified using a percentile-based threshold, and accounts with repeated large debit transactions were flagged. Accounts with zero or negative balances were also identified as overdraft accounts. These criteria helped highlight high-risk accounts requiring closer monitoring.
- Accounts were flagged based on two risk behaviors: frequent large withdrawals and overdrafts. Some accounts showed overdraft risk without frequent large withdrawals, while others may exhibit high spending behavior. This classification helps distinguish different types of financial risk.
- Balance volatility was calculated using standard deviation and coefficient of variation. Standard deviation measures how much an account's balance fluctuates, while the

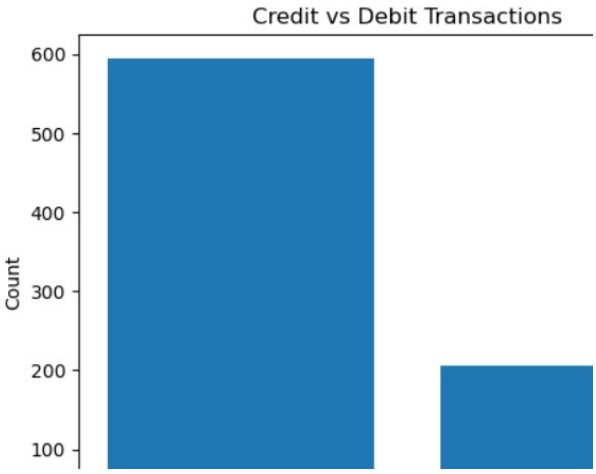
coefficient of variation normalizes this fluctuation relative to the average balance. Higher values indicate unstable account behavior and higher financial risk.

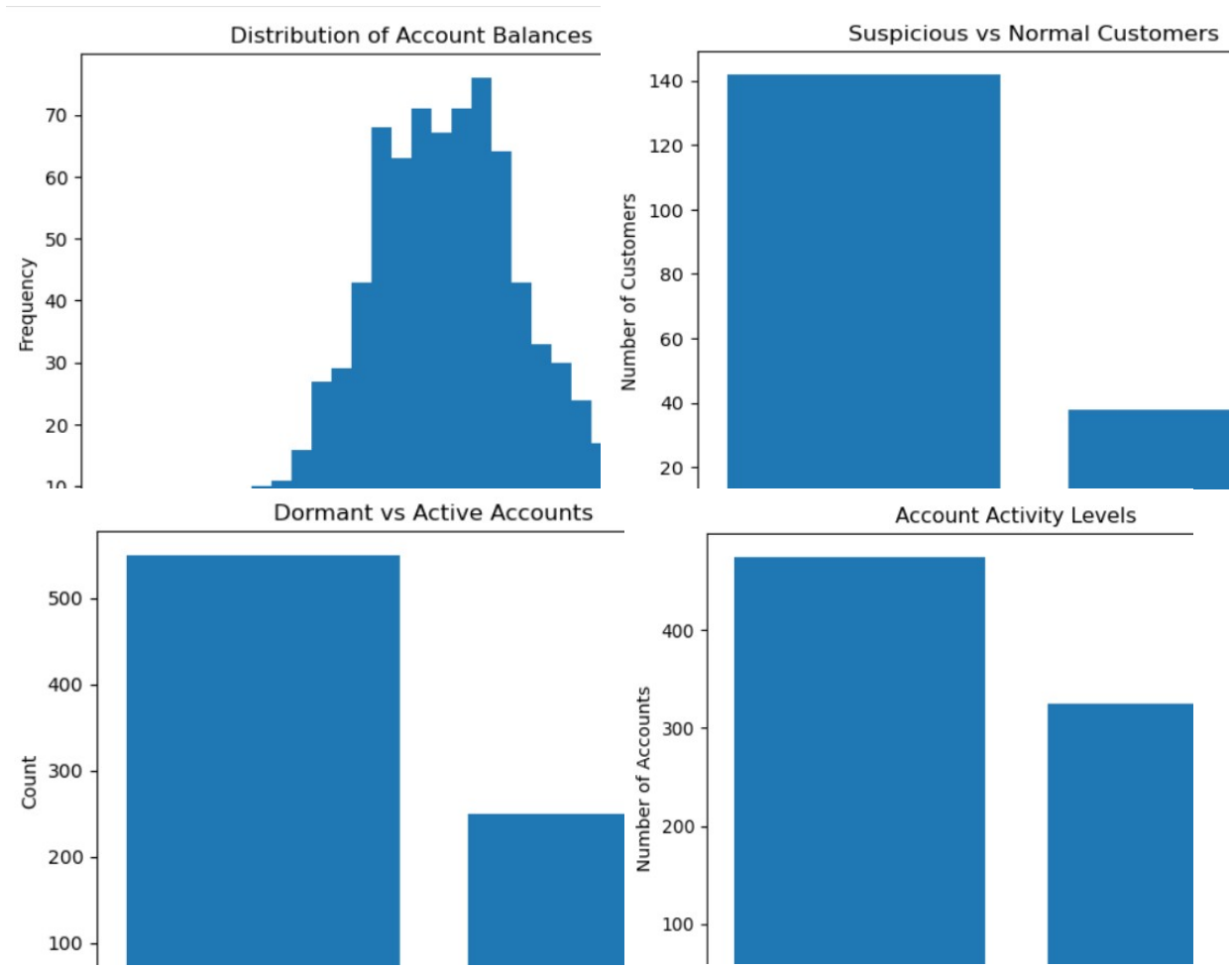
- Anomalies were detected using the IQR method by identifying transactions that fall outside 1.5 times the interquartile range. This approach helps highlight unusually large or small transactions that may indicate abnormal financial behavior. A small number of anomalous transactions were identified. These transactions fall significantly outside the normal transaction range, indicating rare but potentially high-risk or unusual financial activity.
- Based on anomaly detection, withdrawal behavior, balance stability, and volatility metrics, 142 customers were flagged as having irregular or suspicious transaction behavior. These customers exhibited one or more risk indicators and may require closer monitoring. This classification does not indicate fraud but highlights potential financial risk.

Task 5: Visualisation :

- Extensive exploratory data analysis was conducted to understand transaction behavior, balance stability, and financial risk. Visual analysis revealed dominant debit activity, net cash outflows among certain accounts, dormant behavior patterns, and a subset of customers exhibiting irregular or suspicious transaction behavior. These insights support data-driven risk monitoring and customer strategy decisions.







Task 6: Hypothesis Testing :

- An independent t-test was conducted to compare the average balances of high-volume and low-volume transaction accounts. The results showed no statistically significant difference between the two groups ($p = 0.69$), indicating that higher transaction volume does not necessarily correspond to higher account balances.
- Hypothesis testing was conducted to evaluate whether balance volatility differs between medium and low activity segments. An independent t-test showed no statistically significant difference between the two groups ($p = 0.85$). This indicates that account activity level does not significantly impact balance volatility in the dataset.